

Unit II: Current Electricity

CHAPTER-3: CURRENT ELECTRICITY

GIST OF THE CHAPTER:-

Electric current, flow of electric charges in a metallic conductor, drift velocity, mobility and their relation with electric current; Ohm's law, V-I characteristics (linear and non-linear), electrical energy and power, electrical resistivity and conductivity, temperature dependence of resistance, Internal resistance of a cell, potential difference and emf of a cell, combination of cells in series and in parallel, Kirchhoff's rules, Wheatstone bridge.

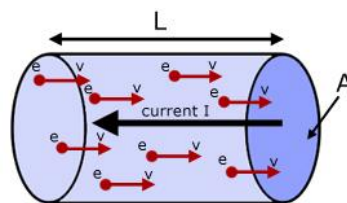
DEFINITIONS & CONCEPTS:-

Electric current- It is the rate of flow of electric charge through a conductor.

Drift velocity- It is the velocity with which a free electron in the conductor gets drifted under the influence of the applied external electric field.

Mathematically -
$$v_d = \frac{e E \tau}{m}$$

Here, A is the area of cross-section of the conductor, τ is average relaxation time and n is number of free electrons per unit volume in the conductor.



Ohm's law- It states that the current flowing through a conductor is directly proportional to the potential difference across its two end provided physical conditions remains same.

Mathematically - $V=IR$, Hence $R=\frac{V}{I}$

Here, R is called resistance of the conductor. $1\text{ohm } (\Omega) = 1 \text{ volt/ampere}$

Ohmic Conductors- Those conductors which obey Ohm's law, are called ohmic conductors e.g., all metallic conductors are ohmic conductor. Examples of ohmic conductors are metals, resistors, nichrome wires, etc.

Non- ohmic conductors- Those conductors which do not obey Ohm's law, are called non-ohmic conductors. e.g., diode valve, triode valve, transistor, vacuum tubes etc.

Resistance -Resistance of a conductor measures the opposition to the current flow. The resistance of a conductor depends on its length l, Area of cross-section A, nature of material and temperature of conductor.

Resistivity or specific resistance- The resistivity of the material of a conductor is defined as the resistance offered by a conductor of this material of unit length and unit area of cross-section. Resistivity is purely a characteristic of the material of wire, it does not depends on length or area of conductor. $\rho = \frac{RA}{l}$

Conductance- The reciprocal of the resistance of a conductor is called its conductance (G). $G = \frac{1}{R}$

Conductivity- The reciprocal of the resistivity of the material of a conductor is called its conductivity.

Mobility-The drift velocity of electron per unit electric field is mobility of electron.

Current Density:- The electric current flowing per unit area of cross-section of conductor is called current density.

It is a vector quantity and its direction is in the direction of motion of positive charge or in the direction of flow of current.

Electric energy- Energy due to electric potential or kinetic energy of charges.

Electric Power:- Power is defined as the rate of absorption/dissipation of energy

Temperature Dependence of Resistivity:

The resistivity of a material is found to be dependent on the temperature.

The resistivity of a metallic conductor is given by $\rho_T = \rho_0 [1 + \alpha(T - T_0)]$

α is called the temperature co-efficient of resistivity.

For metals, α is positive.

For alloys, α change very little with temperature.

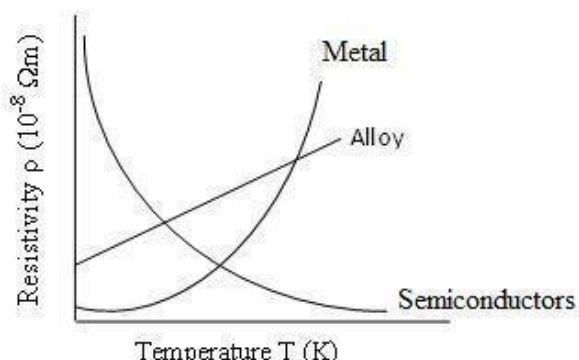
For semiconductors, resistivity decreases with increasing temperature.

$\rho = \frac{1}{\sigma} = \frac{m}{ne^2\tau}$, from this equation ρ depends

inversely on the number n of free electrons per unit volume and on the average time τ between collisions.

With increase in temperature

Conductors	$\tau \downarrow$	$\rho \uparrow$
Semiconductors	$\tau \uparrow$	$\rho \downarrow$



Electrical Energy: The total electric work done.

Energy supplied by the source of emf in

maintaining the current in an electric circuit for a given time

Electric energy = $E = VIt = I^2 R t = \text{Heat}$

Power: The rate at which electric work is done. $\text{Power} = \frac{\text{Work}}{\text{time}} = VI = I^2 R = \frac{V^2}{R}$

Cell, EMF, Internal Resistance:

Cell: 1] Dry Cell 2] Electrolytic cell

EMF: It is the maximum potential difference between the two electrodes of the cell when no current is drawn from the cell.

Internal Resistance: The opposition offered by the electrolyte and electrodes of a cell to the flow of current through it.

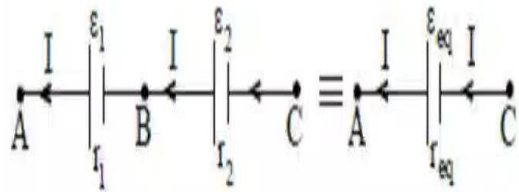
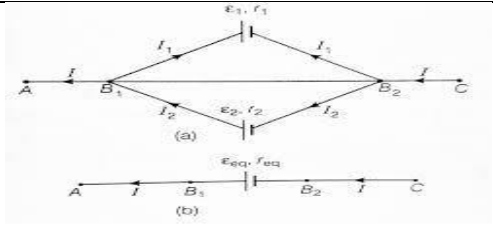
Internal resistance depends on separation between electrodes of the cell, conductivity of the electrolyte, nature of electrodes and common area of the electrodes dipped in the electrolyte.

$$V = \mathcal{E} - Ir$$

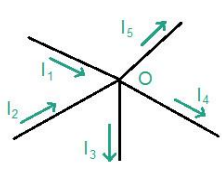
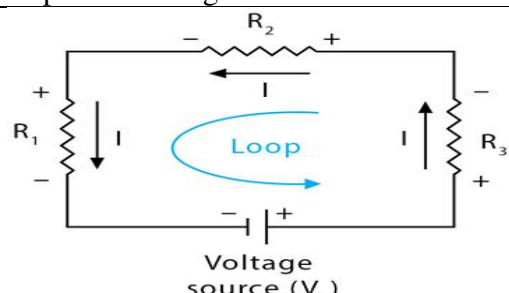
$$V = IR$$

$$I = \frac{\mathcal{E}}{R + r}$$

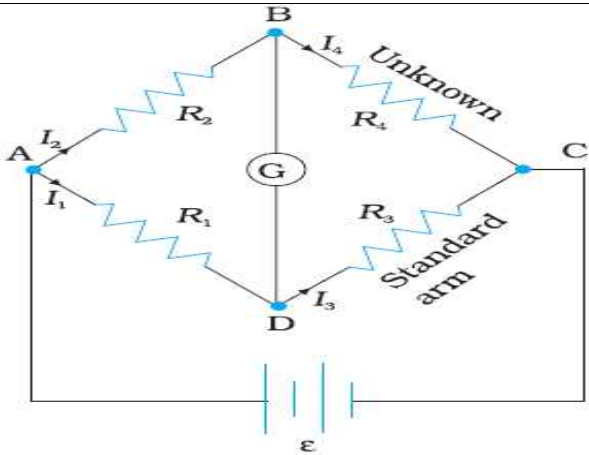
Grouping of cell:

Series Combination	Parallel Combination
	
$\mathcal{E}_{eq} = \mathcal{E}_1 + \mathcal{E}_2$ $r_{eq} = r_1 + r_2$ $I = \frac{nE}{(R + nr)}$	$V = \frac{\mathcal{E}_1 r_2 + \mathcal{E}_2 r_1}{r_1 + r_2} - I \frac{r_1 r_2}{r_1 + r_2}$ $\mathcal{E}_{eq} = \frac{\mathcal{E}_1 r_2 + \mathcal{E}_2 r_1}{r_1 + r_2} \text{ and } r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$ $\frac{\mathcal{E}_{eq}}{r_{eq}} = \frac{\mathcal{E}_1}{r_1} + \frac{\mathcal{E}_2}{r_2} + \dots + \frac{\mathcal{E}_n}{r_n} \text{ and}$ $I = \frac{mE}{(mR + r)}$

Kirchhoff's Rules:

Junction Rule/Current Rule	Loop Rule/Voltage Rule
 According to KCL:- $\Sigma I = 0$ Applying KCL:- $I_1 + I_2 - I_3 - I_4 - I_5 = 0$ $I_1 + I_2 = I_3 + I_4 + I_5$	 Voltage source (V_s) $IR_1 + IR_2 + IR_3 - V_s = 0$
Algebraic sum of all current at junction is zero.	Algebraic sum of changes in potential and cells in the loop is zero.

Wheatstone Bridge:

Wheatstone Bridge	
	
In case of balancing bridge $IG=0$, then $I_1=I_3$ and $I_2=I_4$ so $\frac{R_1}{R_2} = \frac{R_3}{R_4}$	

UNITS AND DIMENSIONS-

SN	Quantity and Units	Dimension
1	Current- Ampere (A) or coulomb /second(C s-1)	[A]
2	Drift velocity- ms^{-1}	$[\text{LT}^{-1}]$
3	Resistance- ohm or volt amp^{-1}	$[\text{ML}^2\text{T}^{-3}\text{A}^{-2}]$
4	Resistivity- ohm-mt	$[\text{ML}^3\text{T}^{-3}\text{A}^{-2}]$
5	Conductance- mho or siemen	$[\text{M}^{-1}\text{L}^{-2}\text{T}^3\text{A}^2]$
6	Conductivity- mho- mt^{-1} or siemen /m	$[\text{M}^{-1}\text{L}^{-2}\text{T}^3\text{A}^2]$
7	Current density- Amp metre^{-2}	$[\text{L}^{-2}\text{A}]$
8	Mobility- $\text{m}^2\text{s}^{-1}\text{V}^{-1}$	$[\text{M}^{-1}\text{T}^2\text{A}]$
9	Energy- Joule or watt-sec	$[[\text{ML}^2\text{T}^{-2}]$
10	Power- Watt or volt-amp	$[\text{ML}^2\text{T}^{-3}]$

FORMULAE-

Physical Quantity	FORMULA	MEANING OF NOTATIONS USED	UNIT
Current-	$I = \frac{q}{t} = \frac{ne}{t}$	q- charge flown in time t n- number of electrons	1 ampere (A) = 1 coulomb /second(C s ⁻¹)
	For non-uniform flow, $I = \frac{dq}{dt}$ Or, $q = \int I dt$		
Drift velocity	$v_d = \frac{e E \tau}{m}$	A-the area of cross-section of the conductor, τ is average relaxation time and n is number of free electrons per unit volume in the conductor.	ms ⁻¹
	$v_d = \frac{eV}{ml} \tau$	Here V is the potential difference between the ends of conductor of length l.	
Ohm's law-	$V=IR$, Hence $R=\frac{V}{I}$	Here, R is called resistance of the conductor.	1ohm (Ω) = 1 volt/ampere
Resistance-	$R = \frac{\rho l}{A}$ $R = \frac{ml}{Ane^2 \tau}$	Here, ρ is resistivity of the material of the conductor.	1ohm (Ω) = 1 volt/ampere
Resistivity-	$\rho = \frac{RA}{l}$ Also $\rho = \frac{m}{ne^2 \tau}$	A-the area of cross-section of the conductor, τ is average relaxation time and n is number of free electrons per unit volume in the conductor.	ohm-meter (Ωm)
Conductance -	$G = \frac{1}{R}$ Also $= \frac{Ane^2 \tau}{ml}$	It is the reciprocal of the resistivity of the material of a conductor	metre ⁻¹ ohm ⁻¹
Conductivity -	$\sigma = \frac{1}{\rho}$ Also $\sigma = \frac{ne^2 \tau}{m}$		
Mobility	$(\mu) = \frac{v_d}{E} = \frac{e \tau}{m}$		
Current Density -	$J = \frac{I}{A} = nev_d$	Relation between current density electric field: $J = \sigma E = \frac{ne^2 \tau}{m} E$	
Resistivity of metallic conductor	$\rho_T = \rho_0 [1 + \alpha(T - T_0)]$	ρ_T =Resistivity at T temperature ρ_0 = Resistivity at reference temperature α = temperature co-efficient of resistivity	$\rho : \Omega m$ $\alpha : ^\circ C^{-1}$ $T : ^\circ C$

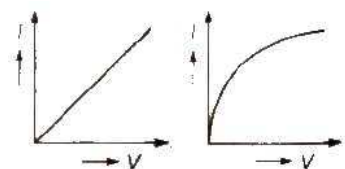
Table 2

Physical Quantity	FORMULA	MEANING OF NOTATIONS USED	UNIT
temperature co-efficient of resistivity	$\alpha = \frac{R_2 - R_1}{R_1(T_1 - T_2)}$	α = temperature co-efficient of resistivity R_2 = Resistance at final temperature R_1 = Resistance at initial temperature T_1 and T_2 = Initial and final temperature	$\alpha : ^\circ\text{C}^{-1}$ $R_1, R_2: \Omega$ $T_1, T_2 : ^\circ\text{C}$
Electrical Energy	$E = VIt = I^2 R t$	E= Energy, V= Voltage, I= Current, R= Resistance, t= Time	E: joule V: volt I: ampere
Power	$P = VI = I^2 R = \frac{V^2}{R}$	P= Power, V= Voltage, I= Current, R= Resistance	P: watt
EMF of Cell	$\varepsilon = \frac{W}{q}$	ε = emf of cell W= work done q= charge	ε : volt W: joule q: coulomb
Potential Difference of Cell	$V = IR = \varepsilon - Ir$	V= Potential difference, I= Current, R= Resistance, ε = emf, r = Internal resistance	V: volt ε : volt r: Ω
Internal resistance of cell	$r = \frac{\varepsilon}{I} - R$	r= Internal resistance, ε = emf, I= Current, R= External resistance	r, R: Ω
	$I = \frac{\varepsilon}{R + r}$		I: ampere (A)
Combination of Cell	1] In Series: $\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2$ $r_{eq} = r_1 + r_2$	ε_{eq} = Equivalent emf r_{eq} = Equivalent internal resistance	ε_{eq} : volt $r_{eq} = \Omega$
	$I = \frac{nE}{(R + nr)}$	I= Current, n= no. cells in series, E= emf, R= external resistance, r= internal resistance	I: A E: volt R,r: Ω
	2] In Parallel: $V = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2} - I \frac{r_1 r_2}{r_1 + r_2}$	V= Potential Difference, I= Current, $\varepsilon_1, \varepsilon_2$ = emf's of cell 1 and 2 r_1, r_2 = internal resistances of cell 1 and 2	V: volt $\varepsilon_1, \varepsilon_2$: volt r_1, r_2 : Ω
equivalent emf	$\varepsilon_{eq} = \frac{\varepsilon_1 r_2 + \varepsilon_2 r_1}{r_1 + r_2}$	ε_{eq} = equivalent emf, $\varepsilon_1, \varepsilon_2$ = emf's of cell 1 and 2 r_1, r_2 = internal resistances of cell 1 and 2	ε_{eq} : volt $\varepsilon_1, \varepsilon_2$: volt r_1, r_2 : Ω

Physical Quantity	FORMULA	MEANING OF NOTATIONS USED	UNIT
equivalent internal resistance	$r_{eq} = \frac{r_1 r_2}{r_1 + r_2}$	r_{eq} = equivalent resistance r_1, r_2 = internal resistances of cell 1 and 2	r_{eq} : Ω r_1, r_2 : Ω
	$\frac{\mathcal{E}_{eq}}{r_{eq}} = \frac{\mathcal{E}_1}{r_1} + \frac{\mathcal{E}_2}{r_2} + \dots + \frac{\mathcal{E}_n}{r_n}$	$\mathcal{E}_{eq}$ = Equivalent emf r_{eq} = Equivalent internal resistance	$\mathcal{E}_{eq}$: volt r_{eq} = Ω
Current due to combination	$I = \frac{mE}{(mR + r)}$	I= Current, E= emf, R= external resistance, r= internal resistance, m= number of cells connected in parallel	I: ampere E: emf R, r: Ω
Kirchhoff's Rules:	1] Junction Rule: $\sum I = 0$ 2] Loop Rule: $\sum (V - IR) = 0$		
Wheatstone Bridge:	$\frac{R_1}{R_2} = \frac{R_3}{R_4}$	R_1, R_2, R_3, R_4 = Resistances	

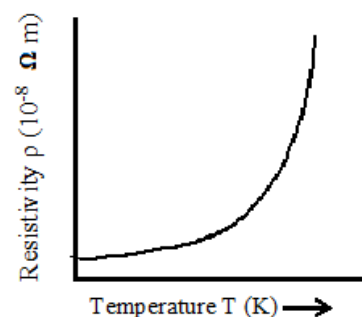
GRAPHS-

Ohmic Conductors

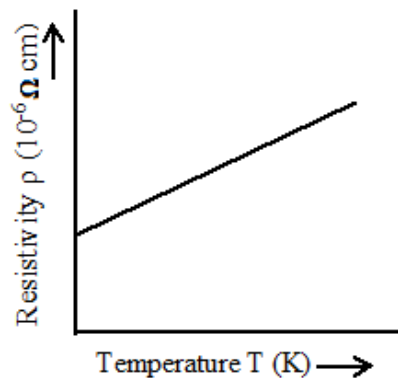


Non-Ohmic conductors

Resistivity ρ of copper as a function of temperature T.



Resistivity ρ of nichrome as a function of temperature T.



Temperature dependence of resistivity for a typical semiconductor.

